

Day 7: Dynamic Risk Scoring Engine (Bayesian Synthesis) & Evidence Vault

Key Milestone: Multi-Vector Bayesian Scoring Synthesis

Sprint Focus: Execution of core deliverables for Day 7 spanning AI/NLP Intent Classification, Email Header Forensics, IOC Intelligence, Relationship Graphing, Case Workflow, and Full-Stack Frontend.

Key Deliverables Completed: Composite Risk Scoring Engine v1.0, Evidence Locker UI, Similar Attacks Recommender.

Individual Team Member Task Logs & Technical Contributions

- Muniswami.Y — AI / NLP Lead

Domain Ownership: Explainable AI, Intent & Phishing Detection, Social Engineering signals | **Key Modules:** NLP/AI Engine, Explainability, Content Risk scoring

Day 7 Execution: Tuned Bayesian weights for NLP indicators: penalty adjustments for spoofed executive display names and fake invoice keywords.
- Jeevan reddy.G — Header & Forensics Lead

Domain Ownership: Email header analysis, SPF/DKIM/DMARC, Mail path reconstruction, Timestamp analysis | **Key Modules:** Header Forensics, Mail Path, Infrastructure Risk

Day 7 Execution: Constructed the Raw Header Forensic Inspector view showing detailed hop-by-hop latency breakdowns and transport encryption (TLS 1.3).
- Lahari.K — IOC & Infrastructure Lead

Domain Ownership: URL, Domain, IP extraction, Geolocation, ASN/ISP mapping, Malicious indicator detection | **Key Modules:** IOC Extraction, Geolocation, URL/Domain/IP Risk

Day 7 Execution: Integrated threat reputation caching to reduce external API latency; implemented domain age and WHOIS registration verification.
- Hemanth.P — Graph & Campaign Intelligence Lead

Domain Ownership: Attack Relationship Graph, Similarity detection, Campaign grouping & correlation | **Key Modules:** Graph Engine, Campaign Detection, Similar Attack Detection

Day 7 Execution: Built Similar Attack Recommendation Engine: surfacing the top 3 historical cases matching the active incident's IOC graph topology.
- Abbas.N — Risk, Case & Response Lead

Domain Ownership: Dynamic Threat Score engine, Automatic Forensic Case creation, Recommended Response actions | **Key Modules:** Risk Scoring, Case Management, Response Recommendations

Day 7 Execution: Developed the Quarantine Vault data models and automated server-side mailbox relocation rule generator.
- Moulaanbee.N — Frontend + Assistant + Simulator Lead

Domain Ownership: Dashboard, Visualizations (Graph, Map, Timeline), AI Chatbot, Email Threat Simulator, Overall UI/UX | **Key Modules:** Dashboard, AI Cybersecurity Assistant, Threat Simulator, Frontend Integration

Day 7 Execution: Created the Evidence Locker modal and interactive Timeline component tracking the lifecycle of forensic artifacts.

System Integration & Quality Verification

Cross-Module Integration: Validated the 0-100 composite risk scoring engine across 150 benchmark test cases (clean, marketing, spear-phishing, BEC).

Quality Assurance & Verification: All technical modules, automated unit test suites, and API contracts for Day 7 executed with 100% clean passes. Code reviewed and merged into main repository.

Prepared By: Threat Trace AI Development Team | **Project Lead:** Muniswami.Y | **Status:** SPRINT DAY 7 MILESTONE SIGNED OFF